

Saehoon Eo

Dept. of Mathematics, Stanford
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Last update: September 14, 2026

EDUCATION

Department of Mathematics, Stanford University

Sep. 2024 – Present

Stanford, CA, United States

- Third-year Ph.D. Student
- Advisor: Prof. Jonathan Luk

Department of Mathematical Sciences, KAIST

Mar. 2020 – Aug. 2024

Daejeon, Republic of Korea

GPA: 4.28/4.3

- B.S. in Mathematical Sciences (Summa Cum Laude, Valedictorian)
- Advisor: Prof. Moon-Jin Kang
- Left for mandatory military service (Jul. 2021 – Jan. 2023)

Korea Science Academy of KAIST

Mar. 2017 – Feb. 2020

Busan, Republic of Korea

GPA: 4.28/4.3

- Government-run gifted high school
- Graduated with honors: Valedictorian, Top Honors in Mathematics

RESEARCH INTEREST

- Partial Differential Equations, Kinetic Theory, General Relativity

AWARDS & HONORS

Dean's List, Department of Mathematical Science, KAIST

Fall 2023, Spring 2023, Spring 2021

Honor Student, Qualification for taking high-level graduate courses

Spring 2021 – Spring 2024

Required to take 85 credits within 6 semesters with GPA higher than 3.7

Korea Presidential Science Scholarship, KSAF

Spring 2020 – Spring 2024

Scholarship for honorable undergrad. in science from the Korean government

KAIST Presidential Fellowship (KPF), KAIST

Spring 2020 – Spring 2024

Granted to 30 undergraduates in about 700 freshmen, join KPF honor society

Graduation Award: Top Prize in Mathematics, KSA of KAIST

Feb. 2020

Distinction Award, Singapore International Mathematics Challenge

May 2018

RESEARCH PUBLICATION

- **Eo, S.** (2026). Nonlinear Phase Mixing in Einstein-Vlasov System near Schwarzschild Spacetime. *Preprint, arXiv:2609.13394*.
- **Eo, S., Eun, N., Kang, M.** (2026). Global Existence of Classical Solutions to Brenner-Navier-Stokes-Fourier System for Large Data. *Preprint, arXiv:2605.07350*.
- **Eo, S., Eun, N.** (2025). Traveling wave solutions to a large class of Brenner-Navier-Stokes-Fourier system. *Comm. Math. Sci.*, 24(6), 1577-1593 (2026)
- **Eo, S., Eun, N., Kang, M., Oh, H.** (2025). Stability of Riemann shocks for isothermal Euler by inviscid limits of global-in-time large Navier-Stokes flows. *Preprint, arXiv:2505.15078*.

- **Eo, S., Eun, N., Kang, M. (2024).** Stability of a Riemann shock in a physical class: from Brenner-Navier-Stokes-Fourier to Euler. *Preprint, arXiv:2411.03613*.
- **Eo, S., Eun, N., Kang, M., Oh, H. (2025).** Traveling wave solutions to Brenner-Navier-Stokes-Fourier system. *J. Differ. Equat.*, 422, 639–658.
- **Eo, S., Hwang, S., Yeo, W. (2020).** Group Rings Satisfying Nil Clean Property. *Communications of the Korean Mathematical Society*, 35(1), 117–124.

ACADEMIC ACTIVITY

PDE Perspectives in Gravitational Theory International Centre for Mathematical Sciences	Sep. 07 – Sep. 11, 2026 <i>Edinburgh, UK</i>
Thematic Program on Shocks and Singularities The Fields Institute	May. 11 – May. 22, 2026 <i>Toronto, ON, Canada</i>
3rd Simons Math Summer Workshop: Partial Differential Equations of Classical Physics Simons Center for Geometry and Physics	Jul. 7 – Jul. 25, 2025 <i>Stony Brook, NY, United States</i>
Workshop on Kinetic Theory and Fluids University of Wisconsin-Madison	Mar. 28 – Mar. 30, 2025 <i>Madison, WI, United States</i>
New Trends in Hyperbolic Conservation Laws and Related Models Courtyard Marriott Namdaemun	Jun. 23 – Jun. 29, 2024 <i>Seoul, Korea</i>
The 10th Korea PDE Winter School Korea Advanced Institute of Science and Technology	Feb. 19 – Feb. 23, 2024 <i>Daejeon, Korea</i>
Fluids in Seoul 2024 June E Huh Center for Mathematical Challenges/KIAS	Jan. 15 – Jan. 19, 2024 <i>Seoul, Korea</i>
IAS/PCMI Undergraduate Summer School Park City Mathematics Institute — Institute for Advanced Study	Jul. 16 – Aug. 5, 2023 <i>Park City, UT, United States</i>
Singapore International Mathematics Challenge NUS High School of Math & Science	May 21 – May 25, 2018 <i>Singapore, Singapore</i>

TEACHING EXPERIENCE

Teaching Assistant for Math 21: Calculus	Winter 2026
Course Assistant for Math 205A: Real Analysis	Fall 2025
Course Assistant for Math 175: Elementary Functional Analysis	Spring 2025
Course Assistant for Math 115: Functions of a Real Variable	Fall 2024
Tutor for Freshmen: MAS101 Calculus I	Spring 2023

TALKS

PDE perspectives in gravitational theory, ICMS	Sep. 8, 2026
Partial Differential Equation Seminar, KAIST	Jul. 22, 2026
Analysis, PDE & Probability Seminar, KIAS	Jul. 13, 2026

OTHER EXPERIENCE

Department's Student Council — Executive Member

Spring 2023 – Spring 2024

- Organize Department Career Seminar for Undergraduates, Fall 2023
- Organize Department Open Lab Event, Fall 2023/Spring 2024
- Mentor for Sophomores, Spring 2023

Republic of Korea Army, 9th div.

Jul. 5, 2021 – Jan. 4, 2023

KAIST Undergraduate Math Colloquium — Member of Operating Committee

Spring 2021

Summer Camp 2019

Jul. 2019

Jacobs University Bremen — VDI

Bremen, Germany

- A week of science camp with international participants
- Financially supported by KSA and Ministry of Science and ICT in South Korea